Statistical disclosure control

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Credits

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Data collection and discosure

- Internet provides unprecedented opportunities for the collection and sharing of privacy-sensitive information from and about users
- Information about users is collected every second
  - Web usage data
  - Social media data
  - Smartphones
  - Wearables
  - …
- Users have very strong concerns about the privacy of their personal information

Data collection and disclosure

- Protecting privacy requires the investigation of many different issues, including the problem of protecting released information against inference and linking attacks
- Huge data collection can now be analyzed by powerful computational techniques and sophisticated algorithms
  - Machine learning
  - Data mining
  - Statistical inference
  - Big Data analytics
  - …

Statistical data dissemination

- Often statistical data (or data for statistical purpose) are released
  - (linked) open data
  - Datasets for experiment reproducibility
  - Data Challenges/Contests/Prizes
  - Policy/Regulations

- Such released data can be used to infer information that was not intended for disclosure

Disclosure

Disclosure can

- occur based on the released data alone
- result from combination of the released data with publicly available information
- be possible only through combination of the released data with detailed external data sources that may or may not be available to the general public
    - New, incredibly rich and detailed external data sources:

When releasing data, the **disclosure risk** from the released data should be very low!

Macrodata vs Microdata

- In the past, data were mainly released in tabular form (**macrodata**) and through statistical databases

- Today, many situations require that the specific stored data themselves, called **microdata**, are released

- Microdata are subject to a greater risk of privacy breaches

Macrodata

Macrodata tables can be classified into the following two groups (types of tables)

- **Count/Frequency**: each cell of the table contains the number of respondents (count) or the percentage of respondents (frequency) that have the same value over all attributes of analysis associated with the table
- **Magnitude data**: each cell of the table contains an aggregate value of a quantity of interest over all attributes of analysis associated with the table

Respondents are businesses, authorities, individual persons, etc, from whom data and associated information are collected for use in compiling statistics.

Count table (example)

Two dimensional table showing the number of beneficiaries by county and size of benefit

	Benefit						
County	**$0-19**	**$20-39**	**$40-59**	**$60-79**	**$80-99**	**$100+**	**Total**
A	2	4	18	20	7	1	52
B	-	-	7	9	-	-	16
C	-	6	30	15	4	-	55
D	-	-	2	-	-	-	2

Magnitude table (example)

Average number of days spent in the hospital by respondents with a desease

	Hypertension	Obesity	Chest pain	Short Breath	Tot
M	2	8.5	23.5	3	37
F	3	30.5	0	5	38.5
Tot	5	39	23.5	8	75.5

Microdata table (example)

- Records about delinquent children in county Alfa

N	Child	County	Educ	Salary	Ethnicity
1	John	Alfa	very high	201	AfrAm
2	Jim	Alfa	high	103	Caucas
3	Sue	Alfa	high	77	AfrAm
4	Pete	Alfa	high	61	Caucas
5	Ramesh	Alfa	medium	72	Caucas
6	Dante	Alfa	low	103	Caucas
7	Virgil	Alfa	low	91	AfrAm
8	Wanda	Alfa	low	84	Caucas

Microdata disclosure risks

- Microdata are subject to a greater risk of privacy breaches

- The main requirements that must be taken into account are
  - identity disclosure protection
  - attribute disclosure protection
  - inference channel

Information disclosure

- Several different definitions of disclosure and different types of disclosure have been proposed
- Disclosure relates to improper attribution of information to a respondent, whether an individual or an organization
- There is disclosure when
  - A respondent is identified from released data (**identity disclosure**)
  - Sensitive information about a respondent is revealed through the released data (**attribute disclosure**)
  - The released data make it possible to determine the value of some characteristics of a respondent more accurately than otherwise would have been possible (**inferential disclosure**)

Identity disclosure

It occurs if a third party can identify a subject or respondent from the released data

Revealing that an individual is a respondent or subject of a data collection may or may not violate confidentiality requirements

- **Macrodata**: revealing identity is generally not a problem, unless the identification leads to disclosing confidential information (attribute disclosure)
- **Microdata**: identification is generally regarded as a problem, since microdata records are detailed
  - identity disclosure usually implies in this case also attribute disclosure

Attribute disclosure

- It occurs when confidential information about a respondent is revealed and can be attributed to it
- It may occur when confidential information is revealed exactly or when it can be closely estimated
- It comprises identification of the respondent and disclosing confidential information pertaining to the respondent

Inferential disclosure

It occurs when information can be inferred with high confidence from **statistical properties** of the released data

- E.g., the data may show a high correlation between income and purchase price of home. As purchase price of home is typically public information, a third party might use this information to infer the income of a respondent

It is difficult to take into consideration this type of disclosure for two reasons

- If disclosure is equivalent to inference, no data could be released
- Inferences are designed to predict aggregate behavior, not individual attributes, and are then often poor predictors of individual data values

Restricted data and restricted access

- The choice of statistical disclosure limitation methods depends on the nature of the data products whose confidentiality must be protected
- Some microdata include explicit identifiers
  - Name
  - Phone number
  - Email address
  - SSN
- Removing such identifiers is a first step in preparing for the release of microdata for which the confidentiality of individual information must be protected

Restricted data and restricted access

The confidentiality of information can be protected by

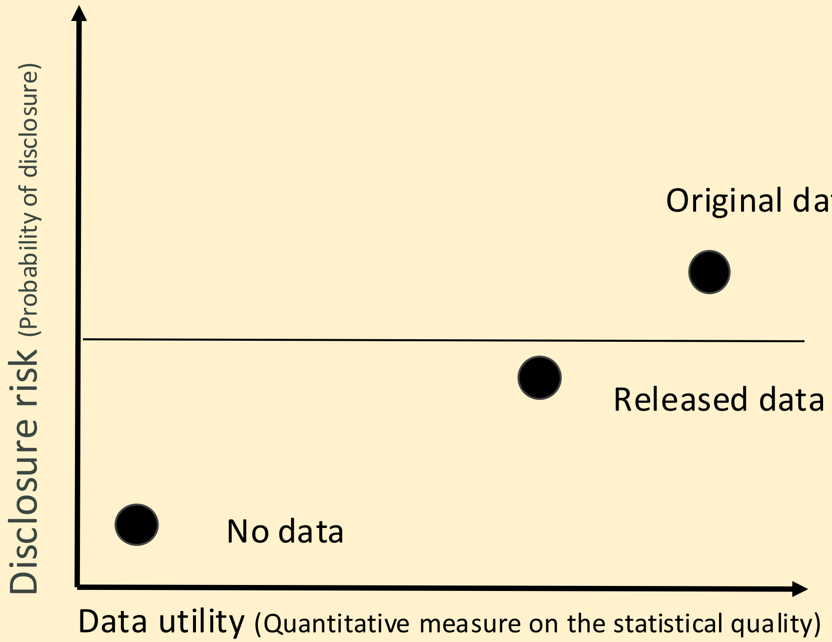
- restricting the amount of information in released tables and microdata (restricted data)
- imposing conditions on access to the data products (restricted access)
- some combination of these two strategies

Statistical disclosure control

Statistical Disclosure Control is a collection of methods that are used as part of anonymization processes to control/limit the risk of re-identification and attribute disclosure through manipulations of the data

- **Disclosure control is not privacy**: all processes which attempt to tackle disclosure risks are primarily technical confidentiality processes which only indirectly and imperfectly map onto privacy protections.

Trade off: utility vs. risk



The higher the disclosure risk is, the higher the data utility is as well.

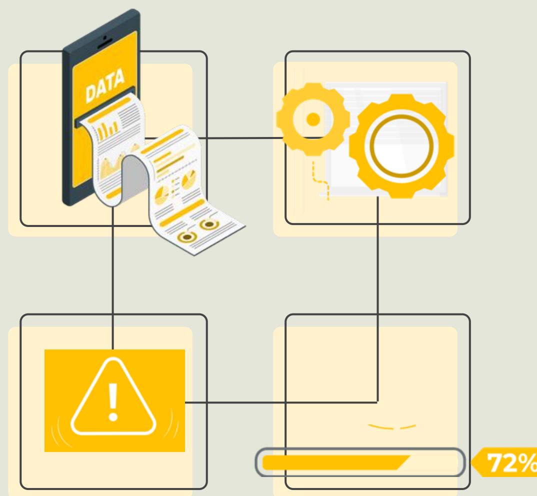
THE TRADE-OFF BETWEEN RISK AND UTILITY ALWAYS EXISTS

COMPROMISE

SDC: procedure

STEP 01

Data



STEP 02

SDC method (+ heuristic parameter choice)

STEP 03

Utility measure

STEP 04

Risk of disclosure is measured

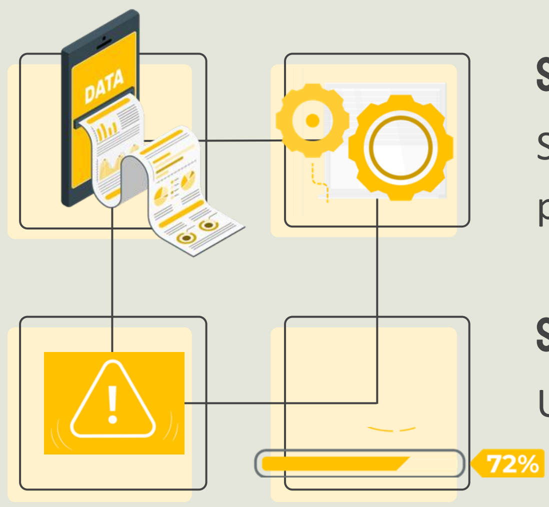
SDC: procedure

STEP 00

Risk analysis

STEP 01

Data



STEP 02

SDC method (+ heuristic parameter choice)

STEP 03

Utility measure

STEP 04

Risk of disclosure is measured

Ex-post

About risk analysis to create micro data that are safe

- From https://ec.europa.eu/eurostat/documents/54610/7779382/Statistical-Disclosure-Control-in-business-statistics.pdf:

The steps to follow to protect business microdata files for researchers, based on already pseudonymised microdata, are:

1. Define quasi-identifiers;
2. Decide on the allowed share of combinations of quasi-identifiers leading to small frequencies (e.g. at most 5% of combinations should lead to low frequencies, e.g. 1 or 2 units – this is an acceptable level of identification risk). The actual level of identification risk will vary, depending on the targeted type of microdata file and data sensitivity.
3. Calculate frequencies for the combinations of quasi-identifiers;
4. Perform a global recode on the quasi-identifiers; /* reducing the number of values of a variable, e.g. by discretisation */
5. Apply SDC methods like micro-aggregation on the variables and records requiring protection (for business microdata, a typical variable to micro-aggregate is turnover).
6. Perform local suppressions on the identifying variables, if there are still records requiring protection;
7. Repeat steps 3-6 until you achieve an acceptable level of identification risk and utility in the file.

SDC Methods

Classified depending on the data format:

Macrodata

Database query outputs

Microdata

SDC methods for macrodata

non-perturbative

Do not modify values in cells

- Cell suppression (CS)

perturbative

Modify values in cells

- Random rounding (RR)
- Controlled rounding (CR)
- Controlled tabular adjustment (CTA)

Cell suppression

In tabular data the cell suppression consists of primary and complementary (secondary) suppression

- **Primary suppression**: withholding the values of all risky cells from publication (e.g., in frequency count tables all cells containing small counts have to be primary suppressed)
- **Complementary (secondary) suppression**: suppress additional non-risky cells to reach the desired protection for risky cells.

Example: healthcare data

Medical Data Released as Anonymous

SSN	Name	Ethnicity	Date Of Birth		Sex	ZIP	Marital Status	Problem
		asian	09/27/64		female	02139	divorced	hypertension
		asian	09/30/64	64	female	02139	divorced	obesity
		asian	04/18/64		male	02139	married	chest pain
		asian	04/15/64	64	male	02139	married	obesity
		black	03/13/63		male	02138	married	hypertension
		black	03/18/63	63	male	02138	married	shortness of breath
		black	09/13/64		female	02141	married	shortness of breath
		black	09/07/64	64	female	02141	married	obesity
		white	05/14/61		male	02138	single	chest pain
		white	05/08/61	61	male	02138	single	obesity
		white	09/15/61		female	02142	widow	shortness of breath

Risky cell

Cell suppression

Ethnicity	Year of birth	Sex	Zip	Marital status	Problem
…	…	…	…	…	…
white	61	male	02138	single	chest pain
white	61	male	02138	single	obesity
white	~~61~~	~~female~~	~~02142~~	~~widow~~	~~shortness of breath~~

Rounding and tabular adjustment

- **Random rounding**: cell values are rounded, but instead of using standard rounding conventions a random decision is made as to whether they will be rounded up or down
- **Controlled rounding**: rounding is performed on each cell, but it is constrained to have the sum of the published entries in each row and column equal to the appropriate published marginal totals
- **Controlled tabular adjustment**: sensitive cell values are replaced by either of their closest safe values and small adjustments are made to other cells to restore the table additivity

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		white	05/14/61	male	02138	single	chest pain
		white	05/08/61	male	02138	single	obesity
		white	~~09/15/61~~	~~female~~	~~02142~~	widow	shortness of breath

64

64

63

64

61

Sensitive values

Year Of Birth

Ethnicity		61	63	64	Total
	Asian	0	0	4	4
	Black	0	2	2	4
	White	3	0	0	3
	Total	3	2	6	11

SDC methods for queryable databases

Query perturbation

Input:

perturbation applied to records

OR

Output:

perturbation applied to query result

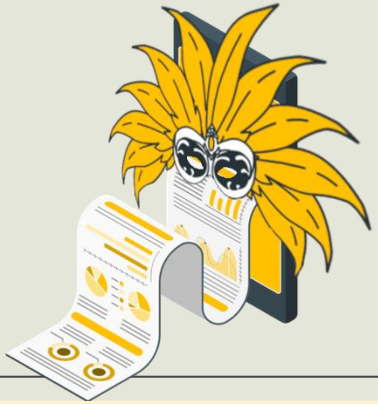
Query restriction

Refuse to answer certain queries:

Need to keep track of previous queries

SDC methods for microdata X

Data Masking



Generates modified version X’

- Perturbative masking
- Non-perturbative masking

Data Synthesis

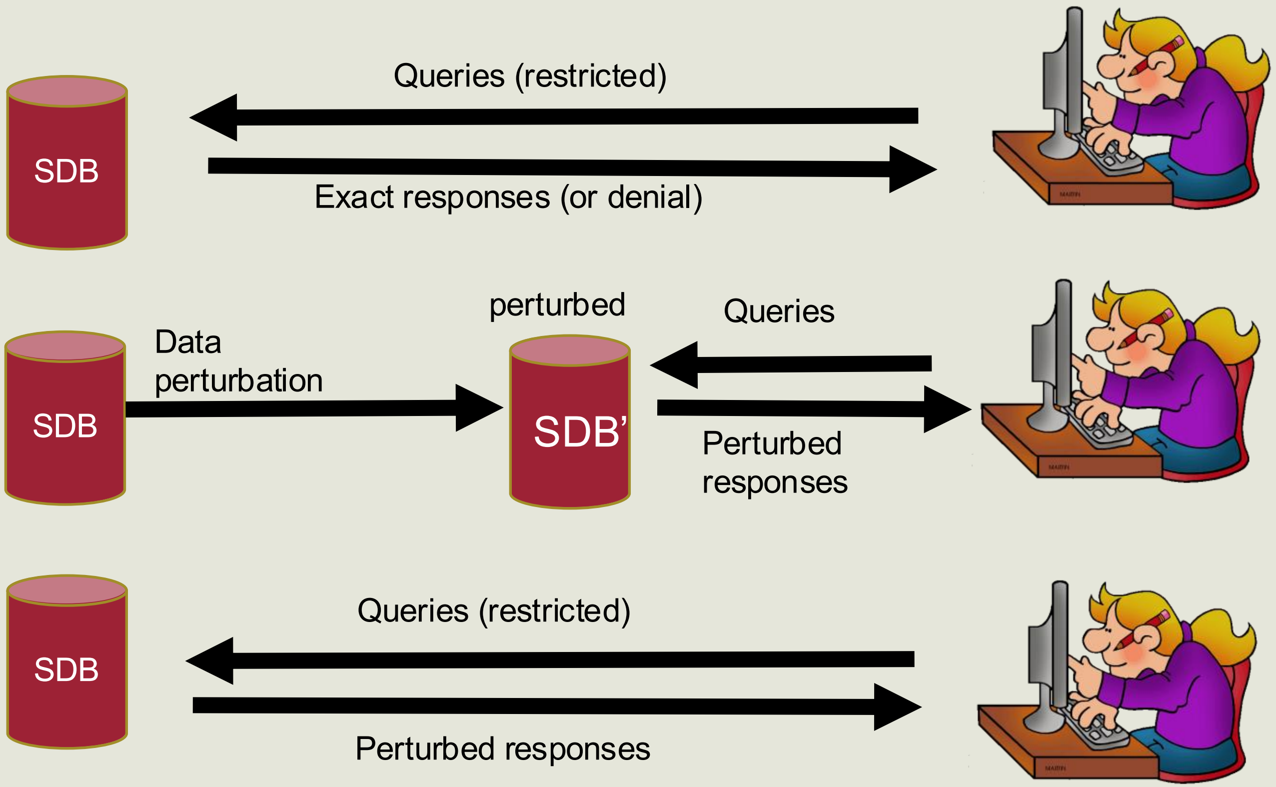
Generates synthetic data X’

X’ preserve pre-selected properties of X

Security to statistical databases

From: ACM Computing Surveys, Vol. 21, No. 4, December 1989

- The main methods to providing security to statistical databases (SDB) against disclosure of confidential information are:



Perturbative masking for microdata

- **Noise addition**
  - Add to each record a noise vector (applicable to numerical microdata)
  - Means and correlations can be preserved
- **Microaggregation**
  - Partitions records into groups (based on within-group similarity)
  - Average record of each group is published
- **Data swapping**
  - Values of each attribute are ranked in ascending order
  - Value swapped with another ranked value randomly chosen within restricted range
- **Post randomization**
  - Works on categorical attributes
  - Confidential attributes changed according to a prescribed stochastic matrix

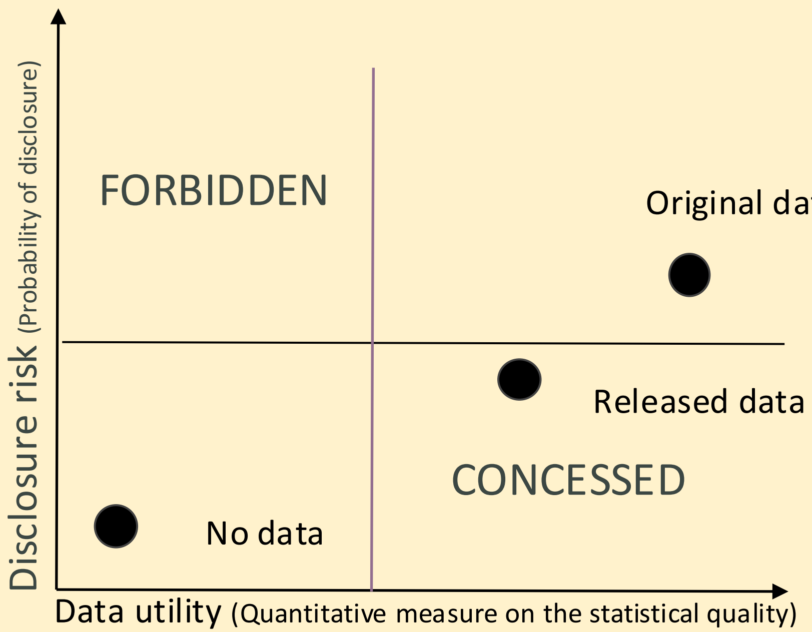
Non-perturbative masking for microdata

- **Sampling**
  - Publishing a sample of original data file
- **Generalization**
  - Categorical attributes are combined in less specific categories
  - Numerical attribute are replaced by intervals
- **Top/bottom coding**
  - Values above/below a threshold are lumped into single top/bottom value
- **Local suppression**
  - Certain individual attributes are suppressed
  - Increase the set of records agreeing on a combination of attributes

Data synthesis for microdata

- **Goal:**
  - release dataset maintaining data subject confidentiality
- **Methods:**
  - synthesize whole dataset (**fully synthetic data**)
  - synthesize only sensitive variables (**partially synthetic data**)
- **Issues:**
  - Synthetic data must yield valid statistical analyses
  - Synthetic data depend on the model used
  - **Overfitting**: synthetic records too similar to original records and re-identification may occur

Trade off: utility vs risk



Try to lower as possible the disclosure risk, with the higher data utility

THE TRADE-OFF BETWEEN RISK AND UTILITY ALWAYS EXISTS

COMPROMISE

See:

Achieving Transparency Report Privacy in Linear Time

https://par.nsf.gov/servlets/purl/10376767

Utility measurement

Utility measurement is difficult because in many cases there is no clarity about what the users of data will want to do with those data.

- **Information loss**: attempts to capture in information-theoretic terms of the change in information caused by disclosure control

Information loss for macrodata

- **Cell suppression**: number or the pooled magnitude of secondary suppressions

- **Rounding and tabular adjustment**: sum of distances between true and perturbed cells (possibly weighted by cell costs if not all cells have the same importance)

Information loss for queries

- **Query perturbation**: the difference between the true query response and the perturbed query response (e.g., it can be characterized in terms of the mean and variance of the added noise - ideally the mean should be zero and the variance small)

- **Query restriction**: the number of refused queries

Information loss for microdata

- **Data use-specific loss measures**: evaluate the extent to which SDC affects the output of particular analyses
- General information loss measures
  - Basic statistics: means, covariances, correlations, …
  - Scores (e.g., Propensity Scores):
    probability of treatment given background variables (*)
  - The distance between the original and disclosure-controlled data using measures (e.g., Jensen-Shannon divergence, Kullback-Leibler divergence…)
    KL divergence is the relative entropy
    JS divergence measures the similarity between two probability distributions (see next slide for details)

(*) Propensity score matching (PSM) is a quasi-experimental method in which **the researcher us**
Using these matches, the researcher can estimate the impact of an intervention.

Details on Kullback-Leibler and Jensen-Shannon divergences

- **Kullback-Leibler** divergence (KL) measures how one probability distribution $P(x)$ is different from a second, reference probability distribution $Q(x)$ with x the value of a random variable X defined on the same space X.
  It is defined as the relative entropy of P given Q:

$$\mathrm{K}L(P \parallel Q) = \sum_{x \in \chi} P(x) \log\left(\frac{P(x)}{Q(x)}\right)$$

- **Jensen-Shannon** divergence (JS) measures the similarity between two probability distributions $P(x)$ and $Q(x)$ using KL. KL is asymmetric; JS is a symmetrized and smoothed version of KL

$$JS(P \parallel Q) = \frac{1}{2} KL(P \parallel M) + \frac{1}{2} KL(Q \parallel M)$$

with

$$M = \frac{1}{2}(P + Q)$$

Ex ante risk analysis

Risk *ex-ante* is based on the notion of **privacy model**, which is a condition, dependent on a parameter, that guarantees an upper bound on the risk of re-identification disclosure and perhaps also on the risk of attribute disclosure by an intruder

- *k*-anonymity
- *l*-diversity
- *t*-closeness
- δ-*presence*
- ….
- differential privacy